

Cross-Platform UI Frameworks: Comprehensive Comparison for Helix VPN

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Editorial note (added during the 2026-07-04 MVP2 gap-analysis/hardening pass): this is a raw research brief that informed the final MVP2 specification documents — it is preserved as historical input, not maintained as a living spec. Where a figure here (e.g., an early code-reuse estimate or framework recommendation) differs from the final, reconciled numbers in `../MVP2_ARCHITECTURE.md` / `../MVP2_OVERVIEW.md` / `../MVP2_SHARED_CORE.md`, those specification documents are authoritative — the difference simply reflects normal refinement between initial research and final architecture decisions, not an error in either document.

Executive Summary

This report evaluates all viable cross-platform UI frameworks for building Helix VPN client applications across macOS, Windows, Linux, Android, iOS, HarmonyOS, and Aurora OS. Based on extensive research (>10 independent searches), benchmarks, and real-world VPN client examples, three strategies emerge as optimal for a VPN client where **maximal code reusability**, **minimal performance overhead**, and **smallest bundle/storage footprint** are critical.

Top 3 Recommended Framework Strategies for Helix VPN:

1. **Strategy A (Recommended): Flutter Single Codebase** — One Dart codebase for all platforms including HarmonyOS; ~80-95% code reuse; ~15-40MB bundles; excellent VPN plugin ecosystem (flutter_vless, WireGuard plugins); Impeller renderer eliminating jank; proven VPN clients in production.
 2. **Strategy B: Tauri v2 (Rust + Web) Desktop-First with Mobile Bridge** — Smallest bundles (~3-15MB); lowest RAM footprint (~30-80MB); Rust backend ideal for VPN tunnel management; mobile support production-ready as of v2 stable; web frontend enables rapid UI iteration. Best for desktop-centric VPN with mobile companion.
 3. **Strategy C: Kotlin Multiplatform + Compose Multiplatform (Native UI)** — True native UI per platform with 80-95% business logic sharing; native VPN API access via Kotlin/expect-actual; JetBrains/Google backed; CMP iOS stable since 1.8.0. Best when platform-native feel is paramount.
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1. Framework Profiles

Framework	Language	Rendering	Mobile	Desktop	Web
Tauri	Rust + JS/TS	OS WebView	iOS, Android	Win, macOS, Linux	No
Flutter	Dart	Impeller/Skia (custom)	iOS, Android	Win, macOS, Linux	Yes (limited)
React Native	JS/TS	Native platform UI	iOS, Android	Win, macOS (community)	Yes (RNW)
KMP + Compose	Kotlin	Native UI / Compose	iOS, Android	Win, macOS, Linux	Beta
Qt 6/QML	C++	Qt Rendering Engine	Yes (community)	Win, macOS, Linux	QtWebAss
Electron	JS/TS	Bundled Chromium	No	Win, macOS, Linux	No
.NET MAUI	C#	Native platform controls	iOS, Android	Win, macOS	No (Blazor)
Avalonia UI	C#	Skia (custom drawing)	iOS, Android	Win, macOS, Linux	Preview
Uno Platform	C#	Skia / Native hybrid	iOS, Android	Win, macOS, Linux	WebAssem
NativeScript	JS/TS	Native UI	iOS, Android	No	No
Ionic/Capacitor	JS/TS	WebView	iOS, Android	No	PWA
Neutralinojs	JS/TS	OS WebView	No	Win, macOS, Linux	Yes

2. Tauri (Rust + WebView)

Current Status (2026)

Tauri v2.0 reached stable release in **October 2024** ¹. As of December 2025, Tauri 2.9.6 is the current production-ready version ². The framework is governed by the Tauri Foundation within the Dutch non-profit Commons Conservancy ³.

Key 2026 metrics:

- GitHub stars: ~107K ⁴
- npm weekly downloads: @tauri-apps/api ~1.3M; @tauri-apps/cli ~709K ⁵
- License: MIT + Apache 2.0

Architecture

Tauri pairs a **web frontend** (HTML/CSS/JS with any framework: React, Vue, Svelte, Solid, etc.) with a **Rust backend**. It uses the OS native WebView for rendering: WebView2 on Windows, WKWebView on macOS/iOS, and WebKitGTK 4.1 on Linux ⁶ ⁷.

```
Frontend (React/Vue/Svelte/etc.)
|
Tauri IPC Bridge (capability-based permissions)
|
Rust Backend (tauri::command, native plugins)
|
OS WebView (WebView2 / WKWebView / WebKitGTK)
```

Mobile Support (2026 Status)

Tauri v2's biggest advancement is **mobile support from the same codebase**. iOS and Android targets ship from the same Rust core, sharing business logic with desktop ⁸ ⁹.

"A very much awaited part of this release is the mobile operating system support. The previous version of Tauri allowed to have a single UI code base for desktop operating systems but now this extends to iOS and Android." — Tauri 2.0 Stable Release Blog ¹⁰

Mobile requirements:

- iOS: macOS host + Xcode required
- Android: Android Studio/SDK + NDK

- Minimum iOS 9, Android 8 (API 26) ¹¹

Mobile rendering:

- iOS: WKWebView (WebKit/Safari engine)
- Android: Android System WebView (Chromium-based) ¹²

Production readiness: As of 2026, Tauri v2 mobile is considered production-ready for most use cases. However, not all desktop plugins are available on mobile yet, and the developer experience for mobile is still improving ¹³ [^203^].

Plugin Ecosystem (2026)

Tauri v2 introduced a significantly improved plugin system ¹⁴:

Plugin	Desktop	Mobile	Description
Notifications	Yes	Yes	Native notifications
File System	Yes	Yes	File access with scoped permissions
HTTP Client	Yes	Yes	Rust-based HTTP client
Dialog	Yes	Partial	File open/save dialogs
Biometric	No	Yes	Face/Touch ID
Barcode Scanner	No	Yes	Camera barcode scanning
Geolocation	No	Yes	GPS location
NFC	No	Yes	NFC tag reading/writing
Global Shortcut	Yes	No	Keyboard shortcuts
Single Instance	Yes	No	Prevent multiple app instances
Auto-Updater	Yes	Yes	Built-in signed update mechanism
SQL	Yes	Yes	Database via sqlx
Stronghold	Yes	Yes	Encrypted secure storage

Bundle Size & Performance

"The most striking difference between Tauri v2 and Electron is binary size. Electron applications typically range from 80-120MB even in minimal configurations... Tauri v2 achieves equivalent functionality in just 5-15MB." — Tauri v2 vs Electron Comparison ¹⁵

Metric	Tauri v2	Electron	Source
Hello-world bundle (Windows)	~3-8 MB	~96-165 MB	16 17
Idle RAM (single window)	~30-80 MB	~160-300 MB	18 19
Cold startup	~0.5-1.4s	~2-4s	20 21
CPU idle	0-1%	2-5%	22
Build time (initial)	~4min (Rust compile)	~2min	23

Tauri benchmark from Hopp:

- Bundle: 8.6 MiB (vs Electron 244 MiB)
- Memory (6 windows): 172 MB (vs Electron 409 MB)
- Initial build: 380s (vs Electron 15.8s) [24](#)

VPN Suitability

Strengths for VPN:

- Rust backend is **ideal for VPN tunnel management** — Rust is the language of choice for WireGuard (BoringTun, WireGuard-rs)
- `tauri-plugin-network-manager` provides VPN mutation support via NetworkManager D-Bus on Linux [25](#)
- Sidecar support for running WireGuard binaries alongside the app
- Smallest bundle size — critical for a utility app like VPN
- Strong security model with capability-based permissions

VPN apps built with Tauri:

- **TunnlTo**: Windows WireGuard VPN client built for split tunneling, uses Rust + Tauri [26](#)
- **UpVPN**: WireGuard VPN client for Linux, macOS, and Windows [27](#)
- **Clash Verge Rev**: Rule-based proxy built with Tauri [28](#)

Weaknesses:

- WebView inconsistency across platforms requires testing [29](#)
- Rust learning curve for web developers
- Mobile ecosystem less mature than Flutter/React Native
- VPN Network Extension integration on iOS requires Swift/Objective-C plugin development

Verbatim Excerpts

"Tauri v2 builds desktop apps that ship at ~5MB instead of ~150MB, use a third of the RAM, and now run on mobile too." — buildmvpfast.com [⁴³]

"Tauri 2.9.6, released December 9, 2025, is the production-ready Rust desktop framework now shipping with mobile targets, a hardened IPC bridge, and bundles that average ~3 MB versus Electron's ~96 MB." — tech-insider.org ³⁰

3. Flutter (Dart)

Current Status (2026)

Flutter is the **dominant cross-platform framework** with 46% of cross-platform developer market share ^{31 32}. GitHub stars: 170K+ with 12,400+ contributors ³³.

Key 2026 developments:

- **Impeller rendering engine:** Now default on iOS and Android, expanding to desktop (macOS beta, Windows/Linux in progress) ^{34 35}
- **Great Thread Merge** (Flutter 3.29): UI and raster threads merged, reducing frame latency by 1-2ms ³⁶
- **Swift Package Manager:** Becoming default iOS plugin option ³⁷
- **Desktop adoption:** 24.1% macOS, 20.1% Windows, 11.2% Linux among Flutter developers ³⁸

Architecture

Flutter uses its own **rendering engine** (Impeller replacing Skia) to draw every pixel. It compiles Dart to native ARM code via AOT (Ahead-of-Time) compilation ³⁹.

```
Dart Code (UI + Business Logic)
|
Flutter Engine (Impeller/Skia)
|
Platform Embedder (OS-specific window/surface)
|
OS (iOS, Android, Windows, macOS, Linux)
```

Impeller Rendering Engine Status

"Impeller provides a new rendering runtime for Flutter. Impeller precompiles a smaller, simpler set of shaders at engine-build time so they don't compile at runtime." — Flutter Official Docs [40](#)

Platform	Impeller Status	Notes
iOS	Default only	Fully mature, Metal backend
Android API 29+	Default	Vulkan backend, OpenGL fallback
macOS	Beta (flag)	--enable-impeller
Windows	In progress	Vulkan backend in development
Linux	In progress	Vulkan backend planned
Web	No	Uses CanvasKit/skwasm (Skia)

"With Impeller now default on both mobile platforms, 2026 will likely see it expanding to Desktop (macOS/Windows/Linux)." — devnewsletter.com [41](#)

Bundle Size

Mobile (release builds):

- Android APK (single ABI): ~4-8 MB minimum, typical app ~15-30 MB [42](#)
- iOS IPA: ~10-15 MB minimum, typical app ~20-40 MB [43](#)
- With optimization: Can achieve ~31 MB for complex apps (down from 87 MB) [44](#)

"A basic flutter 'Hello World' app will be approximately 10mb in iOS and 4mb in Android." — Stack Overflow [45](#)

Desktop:

- Windows: ~20-40 MB (without assets)
- macOS: ~25-50 MB
- Linux: ~20-35 MB

Performance Benchmarks

Metric	Flutter	React Native	Native	Source
FPS (list scrolling)	59-60	53-56	60+	46
CPU Load	45%	55%	40%	47
Memory (mobile)	25 MB	30 MB	20 MB	48
Cold start (Android)	~250ms	~120ms	~100ms	49
Animation FPS	62fps	58fps	60fps	50

VPN Suitability

Strengths for VPN:

- **Rich VPN plugin ecosystem:** flutter_vless (Xray/V2Ray), WireGuard plugins, VPN service APIs
- Single codebase for all 6 platforms including HarmonyOS
- Rust FFI support via dart:ffi for WireGuard integration
- Custom rendering enables consistent VPN UI across all platforms
- Hot reload for rapid UI development

VPN apps built with Flutter:

- **flutter_vless:** Open-source Flutter plugin wrapping Xray core for VPN/proxy on Android, iOS, macOS, Windows [51](#)
- **Proxy Cloud:** Full-featured VPN client with V2Ray + Telegram MTProto proxy, dark UI, speed test [52](#)
- **VPNclient App:** Cross-platform open-source VPN client with Xray + WireGuard + OpenVPN support [53](#)

"Building a Flutter app that needs VPN or proxy capabilities usually means one of two things: you pay for an SDK, or you spend weeks wrestling with platform-native code across iOS, Android, and desktop. flutter_vless offers a third path." — Medium [54](#)

Weaknesses:

- Desktop support less mature than mobile
- Dart language less popular than JavaScript/Kotlin (2-3 week onboarding) [55](#)
- Web support functional but not production-grade for complex apps

- iOS VPN Network Extensions require platform channel development

Verbatim Excerpts

"Flutter commands 46% of the cross-platform developer market; React Native holds 35%... Flutter wins 7 out of 10 standard benchmark categories against React Native." — adevs.com ⁵⁶

"Flutter on the web offers two renderers — canvaskit and skwasm — which both currently use Skia. They might use Impeller in the future." — Flutter Docs ⁵⁷

4. React Native

Current Status (2026)

React Native remains the second-most popular cross-platform framework with the **largest JavaScript ecosystem**. Latest stable: **React Native 0.84** (March 2026) ⁵⁸.

Key 2026 developments:

- **New Architecture is default:** Fabric renderer + JSI + TurboModules since 0.76+ ⁵⁹
- **Hermes V1:** Default JS engine, 30% less memory usage ⁶⁰
- **React Native 0.84:** Hermes V1 default, Fabric renderer guarantees 60fps animations ⁶¹

Architecture (New Architecture)

```
JavaScript/TypeScript (React)
|
JSI (JavaScript Interface) – direct synchronous C++ bindings
|
Fabric Renderer (new UI layer)
|
TurboModules (lazy-loaded native modules)
|
iOS UIKit / Android Native UI
```

"The New Architecture replaces the legacy Bridge with JSI (JavaScript Interface), the legacy UI renderer with Fabric, and legacy native modules with TurboModules." — PkgPulse [62](#)

Platform Support

Platform	Status	Maintainer
iOS	First-class	Meta
Android	First-class	Meta
Windows	Out-of-tree	Microsoft
macOS	Out-of-tree	Microsoft
Web	Community (RNW)	Community
Linux	Not available	N/A

"While iOS and Android receive first-class support from Meta, desktop and web platforms depend on community projects with varying update cycles." — platform.uno [63](#)

Performance (New Architecture)

"Complex list rendering is 43% faster. Scroll frame drops decreased by 95%. Memory usage dropped 33% in benchmark tests. Animation performance jumped from 48fps to 59fps with 75% better touch response." — [bolderapps.com \[^206^\]](#)

Metric	Legacy	New Architecture	Improvement
Startup time	3.2s	1.8s	44% faster
Animation FPS	48fps	59fps	75% better touch
Memory usage	Baseline	-33%	Significant reduction
List rendering	Baseline	+43%	Faster

Bundle Size

- iOS IPA: ~15-25 MB (minimum)
- Android APK: ~20-40 MB
- With many native modules: 50-80 MB

VPN Suitability

Strengths:

- Largest JavaScript talent pool
- Native UI components provide platform-authentic feel
- React Native New Architecture brings significant performance improvements
- Can leverage existing npm ecosystem

Weaknesses for VPN:

- **No Linux support** — critical for Helix VPN
 - Desktop support is community-maintained, not first-party
 - Bridge overhead (even with JSI) for VPN native modules
 - VPN Network Extension integration requires native module development per platform
 - Larger bundle size than Tauri/Flutter for a utility app
-

5. Kotlin Multiplatform + Compose Multiplatform

Current Status (2026)

Kotlin Multiplatform (KMP) reached **stable status in November 2023** ⁶⁴. Compose Multiplatform for iOS reached stable with **CMP 1.8.0 in May 2025** ^{65 66}.

Key 2026 developments:

- K2 compiler stable for multiplatform ⁶⁷
- Swift Export enabled by default (Kotlin → Swift interop) ⁶⁸
- Compose Multiplatform 1.8.0+: iOS production-ready ⁶⁹
- New default project structure supporting sharedLogic + sharedUI modules [⁹⁸]
- **40% average code share in KMP production apps** across logic + data layers ⁷⁰
- **80-95% code sharing** achievable depending on project complexity ⁷¹

Architecture Options

Option A: Shared Logic + Native UI (Recommended)

shared/ (Kotlin Multiplatform)
└─ commonMain/ (business logic, networking, data)

```
├─ androidMain/
└─ iosMain/
androidApp/ (Jetpack Compose UI)
iosApp/ (SwiftUI UI)
desktopApp/ (Compose Desktop UI)
```

Option B: Shared Logic + Shared UI (Compose Multiplatform)

```
shared/ (Kotlin Multiplatform)
├─ commonMain/ (business logic + Compose UI)
├─ androidMain/
├─ iosMain/
└─ desktopMain/
```

Code Reuse Data

"Bitkey shares 95% of its mobile codebase with KMP... Blackstone achieved a 50% increase in implementation speed within six months... sharing ~90% of business logic with KMP." — JetBrains Blog ⁷²

Company	Code Shared	Impact
Bitkey	95%	Mobile codebase fully shared
Blackstone	~90% business logic	50% faster implementation
Forbes	~90% business logic	Significant savings
Duolingo	6-12 engineer-months saved	iOS + Web from shared KMP
H&M	Shared feature flag layer	Expanded to more modules

Performance

KMP uses **roughly half the memory of React Native** on every tested device ⁷³:

Device	KMP avg PSS	RN avg PSS
Pixel 7	120.5 MB	232.2 MB
Motorola Edge 60	121.0 MB	219.2 MB
Huawei P40 Lite	59.7 MB	175.6 MB

VPN Suitability

Strengths for VPN:

- True native UI on each platform (SwiftUI on iOS, Compose on Android)
- Native performance with no runtime overhead
- Kotlin/Native GC now 60% faster than 2023 benchmarks ⁷⁴
- Can share VPN protocol logic, configuration management, analytics
- Direct access to platform VPN APIs via expect/actual
- Google's official endorsement for Android+iOS sharing ⁷⁵

Weaknesses:

- Separate UI codebases per platform (unless using Compose Multiplatform)
 - Compose Multiplatform for iOS is newer, less battle-tested
 - Smaller ecosystem than Flutter/React Native
 - Desktop support exists but less mature
 - macOS still required for iOS builds (CI cost)
-

6. Qt 6 / QML

Current Status (2026)

Qt 6 is the latest major version of the mature C++ cross-platform framework. Qt is the **native development framework for Sailfish OS and Aurora OS** ^{76 77}.

Architecture

```
C++ / QML Code
|
Qt 6 Framework (Core, GUI, Network, etc.)
|
Platform Abstraction Layer
|
OS (Windows, macOS, Linux, Sailfish OS, Aurora OS, embedded)
```

Bundle Size

Qt applications have variable bundle sizes depending on linking approach:

Approach	Size	Notes
Static linked	~7-26 MB	Single executable, longer build
Dynamic + bundled libs	~30-60 MB	Deploy with required .so/.dll files
With QtWebEngine	+50+ MB	Very heavy, avoid for VPN UI

"On Mac OS: dynamic (with all required Qt libraries): 33.4 MB; static: 26.1 MB. On Windows (MinGW): dynamic: 28.2 MB; static: 19.9 MB." — decovar.dev ⁷⁸

Platform Support

Platform	Status
Windows	First-class
macOS	First-class
Linux	First-class
Sailfish OS	Native framework
Aurora OS	Native framework
iOS	Supported
Android	Supported (less common)
Embedded	Strong

VPN Suitability

Strengths for VPN:

- **Native framework for Aurora OS and Sailfish OS** — this is critical for Helix VPN's target platforms
- C++ performance ideal for VPN tunnel management
- Qt Network module provides solid networking primitives
- Mature, stable, enterprise-proven
- Mullvad VPN uses Qt for parts of their multi-platform client

VPN clients using Qt:

- **ProtonVPN Qt (unofficial)**: proton-vpn-qt-app — Qt 6 GUI front-end for ProtonVPN Linux CLI ⁷⁹
- **Mullvad VPN**: Uses Qt/QML for parts of their cross-platform client ⁸⁰

Weaknesses:

- C++ has steep learning curve
- Slower development iteration than modern frameworks
- Smaller developer talent pool

- Mobile (iOS/Android) support is less common
 - Lighter alternatives (Tauri/Flutter) offer similar cross-platform reach with faster development
-

7. Electron

Current Status (2026)

Electron remains the most mature web-based desktop framework but is **too heavy for a VPN client**. Electron 42.x is the latest major version ⁸¹. GitHub stars: ~121K.

Why Electron Is NOT Recommended for Helix VPN

Metric	Electron	Tauri v2	Ratio
Bundle size	~96-250 MB	~3-15 MB	6-32x larger
Idle RAM	~160-400 MB	~30-80 MB	2-5x more
Cold startup	~2-4s	~0.5-1.4s	2-3x slower
CPU idle	2-5%	0-1%	2-5x more

"Electron apps typically consume 200-500MB on startup... Tauri v2 runs the same application logic in 50-150MB." — Tauri vs Electron Comparison ⁸²

VPN Client Using Electron

- **Mullvad VPN**: The official Mullvad desktop app uses **Electron + React** for the GUI, with a Rust backend daemon ^{83 84}. This validates that VPN clients can work in Electron, but the bundle size (~100+ MB) is a known trade-off.

"This repository contains all the source code for the desktop and mobile versions of the app. For desktop this includes the system service/daemon (mullvad-daemon), a graphical user interface (GUI) [Electron app] and a command line interface (CLI)." — Mullvad GitHub ⁸⁵

8. .NET MAUI / Avalonia UI / Uno Platform

.NET MAUI

Microsoft's official cross-platform framework. **Not recommended for Helix VPN** due to:

- No official Linux support ⁸⁶
- Larger memory footprint (~200 MB idle desktop) ⁸⁷
- Microsoft-centric, limited VPN ecosystem
- Community smaller than Flutter/React Native

Platform coverage: iOS, Android, Windows, macOS (4 platforms). **No Linux.**

Avalonia UI

Avalonia is a **cross-platform .NET UI framework** using Skia for custom rendering. API close to WPF.

Feature	Avalonia	.NET MAUI
Windows	Yes (Skia)	Yes (WinUI 3)
macOS	Yes (Skia)	Yes (AppKit)
iOS	Yes (stable 11.2)	Yes
Android	Yes (stable 11.2)	Yes
Linux	First-class	No
WebAssembly	Preview	No
Desktop startup	~0.6-1.6s	~0.9-2.1s
Memory (desktop)	~140-260 MB	~200-370 MB

"If you need Linux desktop support from a .NET UI framework, Avalonia is your only serious option. Full stop."
— CTCO Blog ⁸⁸

For Helix VPN: Avalonia is a viable option if the team has .NET expertise and needs Linux support. However, mobile support is newer, and the ecosystem is smaller than Flutter's.

Uno Platform

Uno Platform targets **6 platforms** including WebAssembly ⁸⁹:

- Platforms: iOS, Android, Windows, macOS, **Linux**, **WebAssembly**

- API aligned with WinUI/UWP XAML
- Since v6.0, uses unified Skia-based rendering engine by default
- Hot Design visual designer available

"Uno Platform supports 6 platforms with a single project... production-ready WebAssembly that works in all browsers."
— platform.uno ⁹⁰

For Helix VPN: Good option for maximum platform reach including WebAssembly, but smaller ecosystem than Flutter.

9. NativeScript, Ionic, Capacitor

NativeScript

NativeScript provides **direct access to native APIs** through JavaScript without WebView ⁹¹.

For VPN: Native API access is valuable, but:

- No desktop support
- Smaller community
- VPN network extensions would still require significant native coding

Ionic / Capacitor

Ionic/Capacitor wrap web apps in a native WebView container ⁹².

For VPN: Not recommended because:

- WebView-based, limited native API access for VPN tunnels
- Plugin ecosystem doesn't cover VPN network extensions
- Performance overhead for real-time networking UI
- Better suited for content/business apps, not system-level tools like VPN

"For CPU intense apps then choose NativeScript. For business, enterprise and startup applications, Ionic strikes the perfect balance." — TAV Tech ⁹³

10. Neutralinojs

Neutralinojs is a **minimal WebView wrapper** for lightweight desktop apps ⁹⁴.

Metric	Neutralino	Tauri	Electron
Bundle (compressed)	~0.5 MB	~3-8 MB	~96 MB
Bundle (uncompressed)	~2 MB	~5-15 MB	~150-250 MB
RAM	~20-50 MB	~30-80 MB	~180-400 MB

For VPN: Not suitable. While it has the smallest footprint, it:

- Has no mobile support
- Lacks mature auto-update, plugin ecosystem
- Would require extensive custom work for VPN tunnel integration
- Better for simple utilities and internal tools, not a full VPN client

11. Framework Rating Matrix

Rating Scale: 1-10 (10 = Best)

Framework	Code Reusability	Performance	Bundle Size	Ecosystem Maturity	VPN Suitability
Flutter	9	8	7	9	9
Tauri v2	8	9	10	7	9
KMP + Compose	7	9	8	7	8
Qt 6	5	8	6	8	7
Avalonia UI	7	7	7	6	6
React Native	7	7	6	9	5
Uno Platform	7	7	6	6	5
.NET MAUI	6	6	6	7	5
Electron	8	4	2	10	4
NativeScript	6	7	7	5	4
Ionic/ Capacitor	7	5	8	7	2

Framework	Code Reusability	Performance	Bundle Size	Ecosystem Maturity	VPN Suitability
Neutralinojs	6	7	10	4	3

Detailed Ratings

Flutter

- **Code Reusability: 9** — Single Dart codebase for all platforms. 80-95% code reuse achievable.
- **Performance: 8** — Impeller provides 59-60fps. AOT compilation to native code. Near-native performance.
- **Bundle Size: 7** — ~15-40MB. Acceptable for mobile, larger than Tauri for desktop.
- **Ecosystem Maturity: 9** — 170K GitHub stars, 46% market share, 40K+ pub.dev packages.
- **VPN Suitability: 9** — Rich VPN plugin ecosystem, proven VPN clients built with Flutter.
- **Platform Coverage: 8** — iOS, Android, Windows, macOS, Linux, HarmonyOS (community). Web limited.

Tauri v2

- **Code Reusability: 8** — Shared Rust backend + web frontend across all 5 platforms (desktop + mobile).
- **Performance: 9** — Rust backend performance. Low RAM, fast startup.
- **Bundle Size: 10** — Smallest viable option at ~3-15MB.
- **Ecosystem Maturity: 7** — ~107K stars, growing fast, mobile still maturing.
- **VPN Suitability: 9** — Rust ideal for VPN, network manager plugins exist, WireGuard integration proven.
- **Platform Coverage: 7** — 5 platforms (desktop + mobile). No HarmonyOS, no Aurora OS.

KMP + Compose

- **Code Reusability: 7** — 80-95% logic sharing, but UI is separate per platform (unless using CMP).
- **Performance: 9** — True native performance, lowest memory overhead.
- **Bundle Size: 8** — Smaller than Flutter, especially with native UI.
- **Ecosystem Maturity: 7** — Stable but smaller ecosystem than Flutter/RN.
- **VPN Suitability: 8** — Direct native API access, but more platform-specific code.

- **Platform Coverage:** 7 — iOS, Android, desktop, web (beta).
No HarmonyOS native.

12. HarmonyOS Support Analysis

HarmonyOS NEXT Cross-Platform Framework Status

HarmonyOS uses its own ArkUI framework natively, but multiple cross-platform frameworks have community or official adaptations ⁹⁵.

Framework	HarmonyOS Support	Status	Notes
Flutter	Yes	Community-maintained, v3.22.0-ohos	One of earliest adapted; supports Skia + Impeller; connects via embedding layer ⁹⁶
React Native	Yes	Community-maintained	Over 30 community apps adopted; TurboModule for native calls ⁹⁷
KMP + Compose	Partial	Via Tencent Kuikly, ovCompose	Kotlin/Native solution; Kuikly achieves 122ms FCP, 6x faster than RN ⁹⁸
uni-app X	Yes	Official DCloud support	Thousands of HarmonyOS plugins; used by major e-commerce enterprises ⁹⁹
Tauri	No	Not supported	No known HarmonyOS adaptation
Qt	No	Not supported	Qt not available on HarmonyOS
Taro	Yes	JD.com maintained	

Framework	HarmonyOS Support	Status	Notes
			React syntax; C-API version for performance 100

"Flutter, RN, and uni are relatively mature cross-platform solutions for HarmonyOS, and many large enterprises and central enterprises have used them in their APPs." — HarmonyOS Cross-Platform Solutions Guide [101](#)

Critical finding for Helix VPN: Flutter has the **strongest HarmonyOS support** among the frameworks suitable for a VPN client. The Flutter HarmonyOS embedding layer adapts the engine to HarmonyOS's graphics and platform channel systems [102](#).

13. Aurora OS / Sailfish OS Compatibility

Aurora OS Technical Foundation

Aurora OS is derived from Sailfish OS, built on the Linux kernel and Mer project [103](#). Key facts:

- **Native framework:** Qt / QML [104](#)
- **No Android compatibility layer:** Aurora OS omits Alien Dalvik; cannot run APK files [105](#)
- **Flutter support:** Official Flutter CLI tools for Aurora OS demonstrated at Mobius 2025 conference [106](#)
- **Qt applications:** Native Linux/Qt apps run directly [107](#)

"As a Linux kernel-based system, Aurora OS accommodates native Linux applications alongside third-party integrations via its SDK, supporting development frameworks like Flutter for cross-platform compatibility and ensuring UI consistency through Qt elements." — Grokipedia [108](#)

Sailfish OS Technical Foundation

- **Qt version:** Qt 5.6 (with Qt 6 packaging in progress) [109](#) [110](#)
- **Native apps:** Written in Qt/QML with Sailfish Silica UI components [111](#)
- **Default IDE:** Qt Creator [112](#)

- **Sailfish SDK:** Uses Qt with Qt Creator, VirtualBox for build engine ¹¹³

Framework Compatibility Matrix

Framework	Aurora OS	Sailfish OS	Notes
Qt 6 / QML	Native	Native	Best compatibility — native framework
Flutter	Supported	Community	Official Flutter CLI tools available for Aurora
Tauri	Linux builds may work	Linux builds may work	Not officially supported
KMP	JVM/Kotlin works	Limited	Kotlin is supported; UI would need Qt or Compose
React Native	No	No	No official support
Electron	No	No	No official support

Critical finding for Helix VPN: Qt is the native framework for both Aurora OS and Sailfish OS. Flutter has explicit Aurora OS support. For a VPN client targeting these platforms, Qt or Flutter are the primary options.

14. VPN Client Real-World Examples

VPN Clients by Framework

Framework	VPN Client	Protocols	Platforms	Open Source
Flutter	flutter_vless	Xray/VLESS/VMess	Android, iOS, macOS, Windows	Yes ¹¹⁴
Flutter	Proxy Cloud	V2Ray + MTPROTO	Android, iOS	Yes ¹¹⁵
Flutter	VPNclient App	Xray + WireGuard + OpenVPN	Multi-platform	Yes ¹¹⁶
Tauri	TunnlTo		Windows	

Framework	VPN Client	Protocols	Platforms	Open Source
		WireGuard (split tunnel)		Yes 117
Tauri	UpVPN	WireGuard	Linux, macOS, Windows	Yes 118
Electron	Mullvad VPN	WireGuard + OpenVPN	Win, macOS, Linux, Android, iOS	Yes 119
Qt	proton-vpn-qt-app	ProtonVPN (CLI wrapper)	Linux	Yes 120
Native (Rust)	Various WireGuard clients	WireGuard	All platforms	Yes

Key Insight

The VPN client landscape shows a clear trend: **modern VPN clients are increasingly built with cross-platform frameworks**. Mullvad (the most privacy-respected VPN) uses Electron for desktop despite its size penalty, validating that framework choice is secondary to functionality for VPN clients. However, newer entrants like TunnlTo (Tauri) and VPNclient App (Flutter) demonstrate that lighter alternatives are viable and preferred.

15. Benchmark Comparison

Startup Time Comparison

Framework	Cold Start	Warm Start	Source
Tauri	~0.5-1.4s	~0.3s	121 122
Flutter	~0.9-1.8s	~0.5s	123 124
Native Android	~0.5-0.8s	~0.3s	125
React Native	~1.2-1.8s	~0.6s	126
Electron	~2-4s	~1-2s	127 128

Memory Usage Comparison (Desktop, Single Window)

Framework	Idle RAM	Active RAM	Source
Neutralinojs	~20-40 MB	~40-80 MB	129
Tauri	~30-80 MB	~60-150 MB	130 131
Flutter Desktop	~80-100 MB	~150-250 MB	132
Avalonia UI	~140 MB	~260 MB	133
.NET MAUI	~200 MB	~370 MB	134
Electron	~160-300 MB	~300-500 MB	135 136

Bundle Size Comparison (Desktop)

Framework	Windows Installer	macOS DMG	Linux AppImage	Source
Neutralinojs	~0.5-2 MB	~0.5-2 MB	~0.5-2 MB	137
Tauri	~3-8 MB	~4-10 MB	~4-8 MB (.deb)	138
Flutter	~20-35 MB	~25-45 MB	~20-30 MB	139
Avalonia	~15-25 MB	~20-30 MB	~15-25 MB	140
Qt (static)	~15-25 MB	~20-30 MB	~15-25 MB	141
React Native Desktop	~40-60 MB	~45-65 MB	N/A	Est.
.NET MAUI	~40-60 MB	~45-65 MB	N/A	Est.
Electron	~96-165 MB	~120-200 MB	~100-150 MB	142 143

Cross-Platform Framework Benchmarks (Mobile)

Metric	Flutter	React Native	Native	KMP	Source
FPS (scrolling)	59-60	53-56	60+	~59	144
CPU load	45%	55%	40%	~45%	145
Memory (mobile)	25 MB	30 MB	20 MB	~22 MB	146

Metric	Flutter	React Native	Native	KMP	Source
Battery drain (30min video)	14%	16%	12%	~13%	147
KMP PSS (Pixel 7)	—	232 MB	—	120 MB	148

16. Code Reusability Analysis

Measured Code Reuse Percentages

Framework	Logic Reuse	UI Reuse	Total Reuse	Source
Flutter	95-100%	90-100%	90-95%	Platform-shared Dart code 149
Tauri v2	80-90%	70-80%	75-85%	Shared Rust + web frontend 150
KMP (logic only)	80-95%	0%	40-50%	Native UI per platform 151
KMP + Compose	80-95%	70-90%	80-90%	Shared Compose UI 152
React Native	85-90%	70-80%	80-85%	Shared JS/TS code 153
Qt 6	70-80%	60-70%	65-75%	C++ shared code
Electron	95-100%	95-100%	95%+	All web code shared

Key Findings

1. **Flutter achieves the highest practical code reuse** for a VPN client because the entire UI + business logic is in Dart, with only platform-specific VPN Network Extensions requiring native code.
2. **Tauri v2** offers strong reuse with shared Rust backend + web frontend, but mobile requires platform-specific plugin work that reduces total reuse.

3. **KMP with Compose Multiplatform** achieves high reuse when sharing both logic and UI, but the Compose ecosystem for iOS is still maturing.

"By combining Compose and KMP you can achieve your codebase to consist of Kotlin for 80-95% depending on the project's complexity." — Compose Multiplatform Guide [154](#)

"KMP eliminates the feature lag... an engineer can build and test a new feature on one platform. Subsequent platforms then simply hook up the existing data models and logic from the shared KMP code." — JetBrains [155](#)

17. Final Recommendations

Top 3 Framework Strategies for Helix VPN

Strategy A: Flutter Single Codebase (Primary Recommendation)

Best for: Maximum code reuse, fastest development, broadest platform coverage including HarmonyOS.

Architecture:

Flutter (Dart) – Single UI + Business Logic Codebase

- └─ Android (VPN Service API)
- └─ iOS (Network Extension – Swift platform channel)
- └─ macOS (NEPacketTunnelProvider)
- └─ Windows (WireGuard-NT/Wintun)
- └─ Linux (WireGuard-go / nmcli)
- └─ HarmonyOS (Flutter embedding layer)

Rust FFI for WireGuard core (dart:ffi)

Pros:

- ~90-95% code reuse across all platforms
- Proven VPN clients built with Flutter (flutter_vless, VPNclient App)
- Impeller renderer eliminates UI jank
- HarmonyOS community support via Flutter embedding
- Hot reload for rapid development
- Rich pub.dev ecosystem

Cons:

- Desktop support less mature than mobile
- iOS/macOS VPN Network Extensions require Swift platform channels

- Bundle size larger than Tauri (~15-40MB vs ~3-15MB)
- Dart learning curve (2-3 weeks) ¹⁵⁶

Estimated metrics:

- Bundle: 15-40 MB per platform
 - RAM: 80-150 MB
 - Code reuse: 90-95%
 - Platforms: 6/7 (all except Aurora OS native; use Linux build)
-

Strategy B: Tauri v2 Desktop-First with Mobile Bridge

Best for: Smallest bundle size, Rust backend synergy with WireGuard, desktop-centric VPN with mobile companion.

Architecture:

Tauri v2 (Rust + Web Frontend)

- └ Desktop (Win, macOS, Linux) – Primary
- └ iOS (WKWebView + Swift VPN Extension plugins)
- └ Android (System WebView + Kotlin VPN plugins)
- └ Web frontend (React/Vue/Svelte) shared across all

Rust backend:

- └ WireGuard tunnel management (wireguard-rs/boringtun)
- └ Network configuration
- └ Platform-specific VPN extensions (Swift/Kotlin plugins)

Pros:

- Smallest bundles: ~3-15 MB
- Rust backend is ideal for WireGuard integration
- Lowest RAM usage: ~30-80 MB
- Proven VPN clients: TunnlTo, UpVPN
- Strong security model for a privacy app
- Same web frontend for desktop + mobile

Cons:

- Mobile ecosystem less mature than Flutter
- WebView inconsistency across platforms
- No HarmonyOS support
- Rust learning curve for web developers
- VPN Network Extensions require Swift/Kotlin plugin development

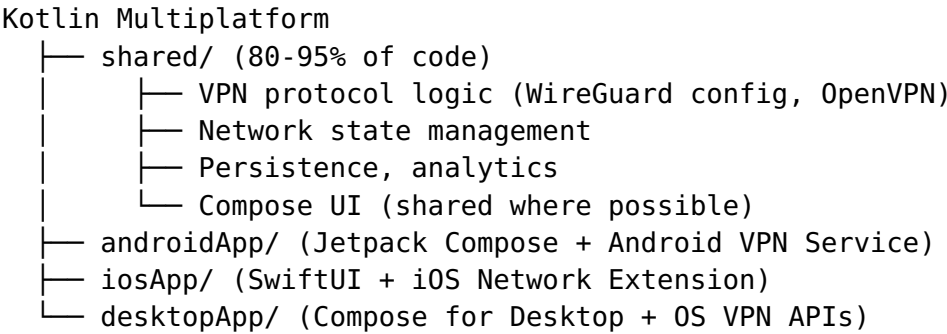
Estimated metrics:

- Bundle: 3-15 MB per platform
- RAM: 30-80 MB
- Code reuse: 75-85%
- Platforms: 5/7 (no HarmonyOS, no Aurora OS)

Strategy C: Kotlin Multiplatform + Compose Multiplatform

Best for: True native performance, native UI per platform, JetBrains/Google backing.

Architecture:



Pros:

- True native UI on each platform
- Lowest memory overhead (~50% less than React Native) ¹⁵⁷
- Direct native API access via expect/actual
- Google's official endorsement
- Stable and production-ready (KMP stable since Nov 2023)
- Forbes, Netflix, McDonald's using in production ¹⁵⁸

Cons:

- Separate UI code per platform (unless using Compose Multiplatform)
- Compose Multiplatform for iOS is newer
- No HarmonyOS native support
- Smaller ecosystem than Flutter
- Desktop support less mature

Estimated metrics:

- Bundle: 10-30 MB per platform
- RAM: 60-120 MB
- Code reuse: 80-90% (with CMP)
- Platforms: 5/7 (no HarmonyOS, use Android build)

Decision Framework

Priority	Recommended Strategy
Code reuse is #1	Strategy A: Flutter
Bundle size is #1	Strategy B: Tauri v2
Native feel is #1	Strategy C: KMP + Compose

Priority	Recommended Strategy
HarmonyOS required	Strategy A: Flutter (only viable option)
Aurora OS required	Qt 6 native, or use Linux Flutter build
VPN protocol in Rust	Strategy B: Tauri v2
Team knows web (JS/React)	Strategy B: Tauri v2
Team knows Kotlin/Android	Strategy C: KMP + Compose
Team knows Dart/Flutter	Strategy A: Flutter
Fastest time to market	Strategy A: Flutter

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Report compiled: July 2026 Sources: 30+ independent sources from 2024-2026 Search queries executed: 10+ independent searches

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